

Project Synopsis

1. Title

Hospital Chatbot for Patient Color-Coded Clinical Pathways

2. Introduction

Hospitals globally, particularly in resource-constrained settings, utilize color-coded triage systems, such as the South African Triage Scale (SATS) or ETAT, to classify patients by severity. These systems typically assign colors like **Red (Emergency)**, **Orange (Very Urgent)**, **Yellow (Urgent)**, and **Green (Non-Urgent)** to guide the patient's care pathway. However, in many settings, delays in this initial triage process can lead to long wait times and poor patient outcomes, as time-sensitive conditions may deteriorate before a human nurse can assess them.

To address this, an AI-powered chatbot can streamline the process by interacting with patients, gathering symptoms, and assigning a color code that indicates the urgency of their condition. This concept mirrors the "Smart Triage" models recently tested in Kenya, which use prediction algorithms to identify high-risk children with greater sensitivity than traditional methods. This chatbot would effectively serve as a "virtual triage nurse" available 24/7, directing patients to the appropriate level of care or department (e.g., Emergency vs. Outpatient) based on their coded clinical pathway, ensuring that the "acuity" of the patient is recognized from the moment they engage with the hospital system.

3. Problem Statement

In many Low- and Middle-Income Countries (LMICs), hospitals struggle with overcrowded waiting rooms and limited triage personnel. For example, in Ghana, physician-to-population ratios can be as low as 0.9 per 10,000, meaning patients might wait for hours in crowded areas before being formally assessed. This delay in triage creates two critical risks:

- **Critical cases are not identified fast enough:** Under-triage (classifying a sick patient as healthy) delays life-saving interventions. Studies show that delays in treatment for high-acuity patients are directly associated with increased mortality.
- **Less urgent cases clog emergency departments:** Patients often lack immediate guidance on where to go. Without an accessible system to identify a patient's triage category (color), "Green" or non-urgent patients congest the Emergency Department instead of moving to outpatient clinics, exacerbating overcrowding and stretching limited resources.

There is an urgent need for an accessible system to quickly identify a patient's triage category

and guide them through the correct clinical pathway, reducing wait times and ensuring urgent cases get prompt attention.

4. Objectives

General Objective:

To develop and evaluate an AI-powered hospital chatbot that assigns triage color codes to patients and navigates them through the appropriate clinical pathways to reduce delays and congestion.

Specific Objectives:

1. **To Develop a Symptom-to-Color Algorithm:** Create a chatbot logic flow based on validated protocols (like SATS/TEWS or Smart Triage prediction models) to calculate urgency colors (Red, Orange, Yellow, Green).
2. **To Map Hospital-Wide Pathways:** Define the specific clinical pathway for each color (e.g., "Yellow patients proceed to Zone B for vitals; Green patients proceed to Pharmacy").
3. **To Evaluate Triage Accuracy:** Compare the chatbot's color assignments against standard nurse triage to assess sensitivity (ability to catch sick patients) and safety.
4. **To Measure Impact on Flow:** Assess if the chatbot reduces the "Time to Provider" and the rate of patients leaving without being seen (LWBS).

5. Methodology

- **Study Design:** A mixed-methods study involving system development and a comparative accuracy assessment.
- **Setting:** A Hospital in an LMIC setting (e.g., Ghana) utilizing standard color-coded triage (SATS or similar).
- **Target Population:** Patients presenting to the hospital (excluding immediate life-threatening emergencies who cannot interact with a device).

Data Collection:

- **Algorithm Validation:** Retrospective testing using patient vitals and outcomes to see if the chatbot predicts the correct color.
- **Usability Testing:** Surveys with patients to determine if the "virtual nurse" guidance is clear and helpful.

6. Significance

By deploying a virtual triage nurse, this project aims to:

- **Decongest the ED:** By accurately routing "Green" patients to non-emergency pathways.

- **Improve Safety:** By flagging "Red/Orange" patients immediately, potentially reducing mortality associated with waiting times.
- **Standardize Care:** Providing consistent, algorithmic assessment 24/7, reducing variability in patient handling.